

Section 8.3: The Design Stage

When analysis (stage 1) has been completed, the **requirements for the new system should be very clear**.

This brings us to the design stage. This is where the new system is **planned thoroughly** to make sure that the developers know how **each screen should look** and how the **system should work**.

Key Concepts of this section:

- # Understand what is **involved** in the **design** stage.
- # Know the **different types of designs** that are produced.
- # Be able to describe **different examples and uses** of data **validation** and data **verification**

Stage 2 - Design

What is design?

- # Once there is a clear list of requirements for the new system it is time to **design (or plan)** how it will **look and work**.
- # It doesn't matter what type of system is being created (Web site, database, payroll etc), it is **very important** that **clear designs are produced**.
- # The designs for the new system have **a lot of thought** put into them to **try and reduce faults/errors at a later stage**.

If the system is **built without proper planning**, there is a good chance that **faults** will appear after system is created.

This would mean that the system would have to be changed in order to correct the faults. This would be very **expensive** and **time consuming**!

What is included within these designs?

- # Every part of the system is carefully designed and tasks include:

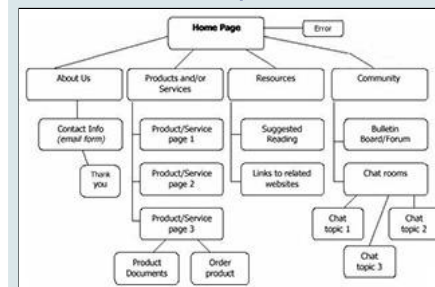
- 1 Designing data entry screens**
(how will data be entered into system.... e.g. text boxes, drop-down menus etc)



Key Words:

Design, Validation Check,
Verification Check

Examples:



Web site layout design showing which pages will link to each other.

- 2 Designing user-interface layouts**
(what will the system's menus, search facilities etc look like?)
- 3 Designing printed outputs (reports etc)**
(what will printed outputs look like.... e.g. payslip's, customer bills etc)
- 4 Designing screen-based outputs**
(what will outputs that are displayed on the screen look like?)
- 5 Designing structures to store data**
(e.g. for databases, tables will need to be designed to hold the system's records)
- 6 Designing data validation methods - (See below)**
(how will the system prevent invalid / incorrect data from being entered?)
- 7 Designing data verification methods - (See below)**
(how will the system check that entered data is correct?)

Data entry screens must be designed to be easy to use. Note the different methods of entering data (text box, check box, drop-down-list). Click image to zoom

Printed reports have to be designed to make sure that they are neat and tidy and easy to read. (click to zoom)

We will now take a closer look at the last two items in the list above.... **validation** and **verification**:

1 Designing System Validation

- # Validation is where a computer system checks that **data entered** by the user is **sensible** and **usable**.
- # The computer system will be **programmed with a list of these checks** that it can use to compare to the data that a user is entering.

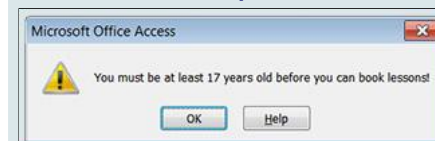
If the user try's to enter something that does not match up with items on the validation check list, the system will not accept the data.

- # These checks are often called '**Validation Rules**'.

An example of a validation rule - driving school booking system

Validation rule in the system states that learner drivers need to be **17 years or older (>=17)**

Examples:



A validation check has shown that a student is too young to book a driving lesson. The system flags up the error.

If a user try's to enter a new student's **age as 16** the system will **not allow the data to be entered** and the person will **not able to book driving lessons**.



Another validation check has picked up an error with a gender field (probably a typing error). The system is instructing the user that they must enter male or female.

Examples of different validation checks:

Validation Check	Description	Example
Presence check	Checks that data has been entered and not missed out.	If a user misses out required data (such as a postcode), the system not allow them to progress.
Length check	Checks that the correct number of characters has been entered.	If a 'date' field requires the user to enter 8 digits (dd/mm/yyyy) , they must enter all 8 or the system will reject.
Range check	Performed on numbers to check they are within certain ranges.	Persons age should be in the range of >0 but <120 .
Limit check	Performed on numbers to make sure that data is not above or below a specified limit.	A learner driver's age must be >=17 .
Format check	Makes sure that data is in a specific format.	Dates should be entered in the format of dd/mm/yyyy .
Data type check	Checks that entered data does not contain invalid characters.	Pure number fields (distance, temperature etc) should not allow for text to be entered.
Consistency check	Makes sure that data in one field is consistent with data in another field.	If ' Male ' has been typed into a gender field, the person's title cannot be 'Mrs' .

2 Designing System Verification

- # Verification is a way of making sure that **data being entered** into the system **exactly matches the source of the data**.

For example:

Imagine that **data about employees (addresses, phone numbers etc)** is being entered into a computer system by typing in information found on a **printed form** that each employee filled in.

The data in the **system** must be **identical to the data on the printed forms** or there could be problems if an employee needs to be contacted at home at a later data.

- # There are two types of verification checks that makes sure that data in the system is accurate:

Examples:

Registering new accounts often require you to enter a password twice. This makes sure that the user makes no accidental typing errors.

Verification Check	Description
Double entry	This is where data is entered twice by two different people. The computer compares the two entries and lets the user know if it finds any differences. NOTE: Double entry is often used when you set up online accounts. You are often asked to entered your choice of password twice to make sure that you haven't pressed an incorrect key by mistake.
Visual check	This is where the person entering data into the computer system carefully compares what they have entered with the data in the original document.

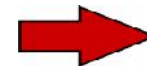


Visual checks of data require the user to carefully check what they have entered into the system against what was on the original document.



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